

KAYCEE NDUKUBA

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PROFESSIONAL SUMMARY

AI / Machine Learning Engineer (MSc) who builds LLM-powered and NLP systems end to end — from data pipelines to deployed, explainable models. Built a portfolio of applied machine learning and AI projects on real-world datasets spanning retrieval-augmented AI, NLP, customer analytics and behavioural modelling, with a focus on translating technical outputs into measurable outcomes. Experienced with production ML workflows (Docker, AWS/GCP, GitHub Actions, MLflow) and grounded, evaluation-driven AI development, backed by a psychology background that keeps the human outcome in view.

CORE SKILLS

AI & LLMs: Large Language Models, Retrieval-Augmented Generation (LangChain, FAISS, sentence embeddings), prompt engineering, evaluation of AI outputs

NLP & Machine Learning: Text classification, sentiment analysis, predictive modelling, feature engineering, model evaluation (ROC-AUC, F1), SHAP interpretability

Programming: Python (pandas, NumPy, scikit-learn, XGBoost, gensim), SQL, R, Git/GitHub

MLOps & Deployment: Docker, GitHub Actions (CI/CD), MLflow, AWS, GCP, ETL pipelines

Foundations: PyTorch / TensorFlow (neural-network coursework), vector search, data cleaning & transformation

EXPERIENCE

AI & Data Scientist — Yves OS, London

Dec 2025 – Present

- Developed behavioural analytics workflows that transformed multi-platform user-interaction data into measurable insights, used to prioritise product features and improve user experience.
- Applied large language models to generate personalised recommendations and summaries, then **evaluated outputs against defined quality criteria** to refine prompt design and improve analytical quality.
- Iteratively tested and refined AI-driven recommendation workflows using user feedback and exploratory analysis, improving recommendation relevance.
- Built engagement and productivity metrics that quantified user behaviour and informed data-driven product decisions, communicated to stakeholders through dashboards and reports.

Python · SQL · Machine Learning · LLMs · pandas · PostgreSQL · AWS

Machine Learning Engineer — Freelance / Contract

Jan 2025 – Dec 2025

- Delivered applied machine learning and analytics across multiple private-sector engagements, translating business questions into models and measurable recommendations.
- Selected engagements** — customer analytics, churn modelling, predictive modelling, demand forecasting and behavioural analysis.
- Built end-to-end analytical workflows — data preparation, feature engineering, model training and evaluation — in Python and SQL.
- Deployed and maintained production models with Docker on AWS/GCP, automating testing and model versioning via GitHub Actions and MLflow.

Python · scikit-learn · SQL · AWS · GCP · Docker · GitHub Actions · MLflow

Data Analyst — Plan-Net, Leicester

Oct 2023 – Nov 2024

- Automated recurring reporting with SQL and Python, reducing manual effort and improving turnaround for business stakeholders.
- Designed and maintained ETL workflows that improved data quality, consistency and reliability across reporting pipelines.
- Partnered with cross-functional stakeholders to translate operational data into clear reports and recommendations that supported decision-making.
- Analysed infrastructure and operational data to surface trends, recurring issues and opportunities for optimisation.

SQL · Python · ETL · dbt · Snowflake · Git

SELECTED PROJECTS (FULL CODE: [GITHUB.COM/KNDUKUBA17-HUB](https://github.com/kndukuba17-hub))

Enterprise RAG Knowledge Assistant · Python, LangChain, FAISS

- Built the retrieval layer of a Retrieval-Augmented Generation system: document chunking, sentence-transformer embeddings and FAISS semantic search over policy documents.
- Designed a document-agnostic pipeline that returns cited source passages for a natural-language query.

Aviation Sentiment Classifier (NLP) · Python, scikit-learn, gensim

- Classified **14,640 real airline tweets**; Bag-of-Words + Logistic Regression (GridSearchCV) reached **80.4% accuracy / 0.74 macro-F1**, beating a Word2Vec model.
- Engineered custom scikit-learn transformers (structural text features + cleaning) inside a FeatureUnion pipeline.

Customer Churn Prediction · Python, scikit-learn, XGBoost, SHAP

- Predicted churn on **7,043 real telecom customers**; Logistic Regression achieved **ROC-AUC 0.842** (5-fold CV 0.845), 0.78 churn recall.
- Optimised for recall/ROC-AUC over accuracy; targeting the top 20% risk reaches **~50% of churners**.
- Handled class imbalance and explained drivers with coefficients and SHAP.

Customer Lifetime Value (CLV) Prediction · Python, XGBoost, SHAP

- Modelled 6-month customer spend from RFM behaviour on **1.07M real transactions** (UCI Online Retail II) with a leakage-safe temporal split.
- XGBoost reached test R^2 0.69 / MAE £600; the **top 10% predicted customers captured ~63% of actual future revenue** (vs 10% at random).
- Ranked the behavioural drivers of value with SHAP to make the output actionable for marketing.

EDUCATION

MSc, Applied Data Science & Cyber Security — Royal Holloway, University of London

2025 – 2026

Modules: Machine Learning, Neural Networks, Big Data Analytics, Statistical Modelling, Cyber Threat Intelligence.

BSc, Psychology — University of Leicester

2021 – 2024

Modules: Quantitative Research Methods, Behavioural Statistics, Cognitive Neuroscience.

A-Levels — Langley Park School

2019 – 2021

Modules: Chemistry (A), Biology (A), Psychology (A).

Hobbies

Sports & Strategy: Fascinated by game theory and the analytical side of high-performance team sports.

Reading & Development: Avid reader of psychology and financial strategy (e.g., *Atomic Habits*, *Rich Dad Poor Dad*).

Music & Fashion: Deeply passionate about the intersection of design, sound, and modern culture; enjoy dissecting industry trends and applying mechanics to personal pursuits.